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VHFACSourceDefTopic

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About the VHFAC Source

About the VHFAC Source

The VHFAC Source instrument can generate high-accuracy, low-noise, and low-distortion signals with a bandwidth of 160 MHz. It has a maximum user-programmable sampling rate of 400Msps and can drive a peak differential output of -4 to +4 V with DC coupling.

This section includes the following topics:

- Features
- Safety Information

VHFAC Source information includes the following sections:

- VHFAC Source Theory of Operation
- VHFAC Source AC Calibration
- VHFAC Source Programming
- VHFAC Source Debug Display
- VHFAC Source Examples

Related Topics

- VHFAC Specifications
- VHFACSource (VBT syntax)
- VHFAC DIB Design Guidelines

For information on other VHFAC instruments see:

- About the VHFAC Capture
- About the VHFAC Serial Bus

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About the VHFAC Source : Features

Features

Source instrument features include:

- High-speed, high-resolution DAC
- DC to 160 MHz analog bandwidth
- Wide filter selection
- Harmonics: -70 dBc @ 10 MHz (-30 to 0 dBm), -60 dBc @ 40 MHz (-30 to 0dBm)
- Spurs: -70 dBm @ 10 MHz (-30 to -10 dBm), -55 dBm @ 40MHz (-30 to -10dBm)
- Large waveform and event memories
- Event lines synced to Source for TV/VTR
- Multiplexed, single-ended or differential outputs

For VHFAC Source hardware specifications, see VHFAC Specifications.

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About the VHFAC Source : Safety Information

Safety Information

When operating the VHFAC Source instrument, always follow general UltraFLEX test system safety guidelines. See Test System Safety for details.

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VHFAC Source AC Calibration

VHFAC Source AC Calibration

The VHFAC Source instrument incorporates an AC run-time calibration to help guarantee amplitude accuracy. Hardware calibration is performed in the factory before shipment. IG-XL applies the calibration factor based on the user program. You must specify both a frequency at which to apply the calibration factor and a mode by which the calibration factor should be applied. You can choose one of three AC run-time calibration modes, which can only be programmed using VBT .Pins language:

None	Calibration is not performed.
Always	Calibration is performed each and every time it is started.
Automatic	Calibration is performed only when a new instrument setup is encountered. This is the default and recommended mode.

You must also program the frequency at which to apply the calibration factors. You do this using the Calibration Frequency parameter, and you can program it using Signals VBT language.

For best amplitude accuracy results with dual-tone and multitone signals, manual calibration is recommended with the Calibration Frequency parameter programmed to a specific tone, oftentimes the highest frequency tone.

VBT Example

The following is example VBT code:

```
' Code to set AC Runtime Calibration mode for VHFAC Source Pin "SrcPin" to Automatic
```

TheHdw.VHFACSource.Pins(“SrcPin”).Calibration.ACRuntime.Mode = tlVHFACSourceACCalibrationModeAutomatic

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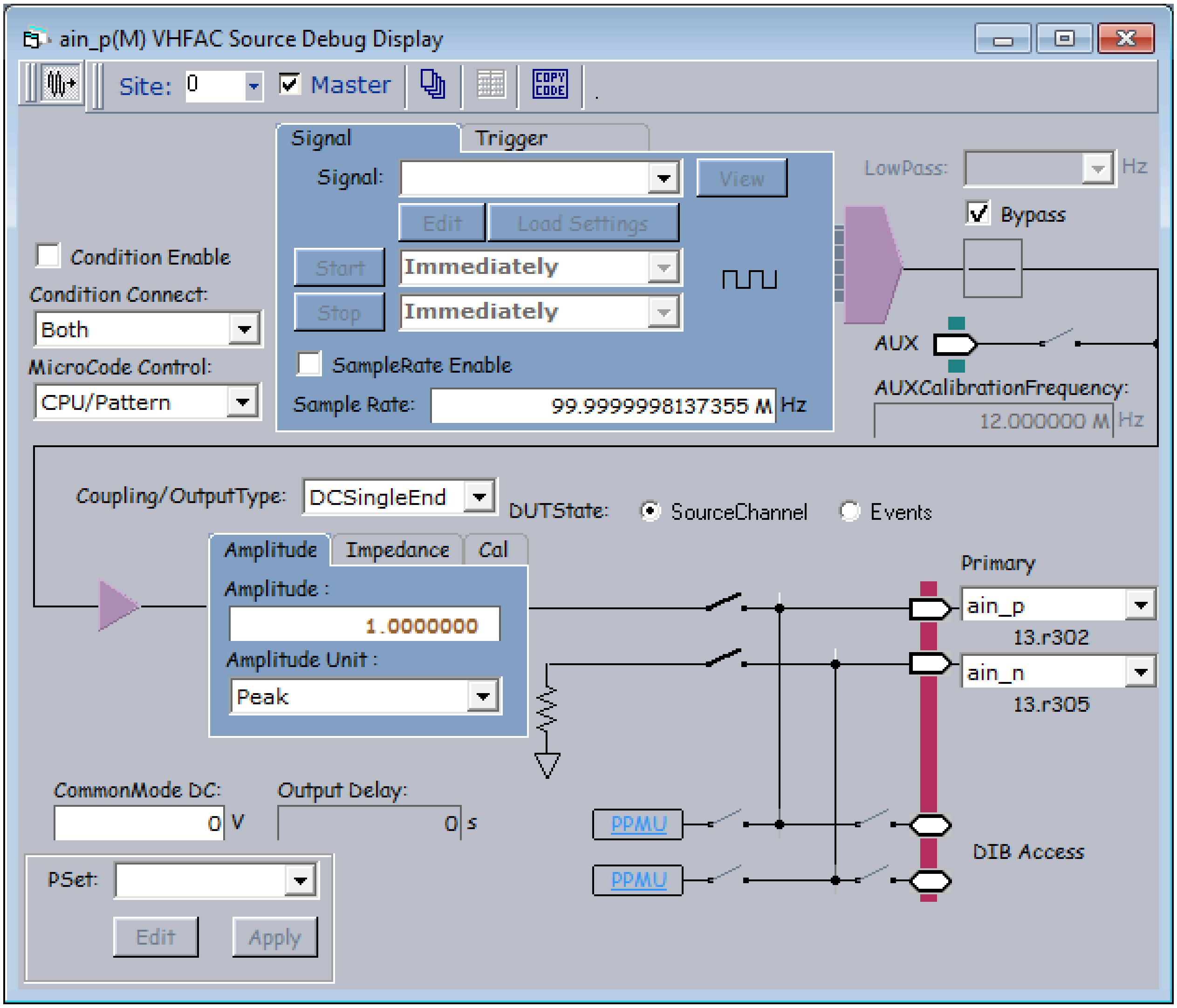
VHFAC Source Debug Display

VHFAC Source Debug Display

The VHFAC Source Debug Display is an interactive tool that shows the state of the VHFAC Source instrument in real time. Most of the parameters are interactive. Features that can be changed may be limited to a certain range or selection of options provided in a drop down menu. Non-interactive features are grayed out on the display and cannot be adjusted during debugging.



VHFAC Source Debug Display

This section describes features specific to the VHFAC Source debug display. For more information on using debug displays, see the following:

•	Using Debug Displays in TDE

•	Producing Code from Debug Displays

Starting the VHFAC Source Debug Display

To launch the VHFAC Source Debug Display, on the IG-XL Tools tab, in the Launch group, select Debug Displays>VHFAC Source.

To start debugging, select a pin and a site to display the programmed values. Select sites and pins from the drop-down menu.

Without a test program loaded, the initial debug display produces and an error message:

No pin exists of type VHFAC

and the connection lines appear directly connected to the DUT (without relays, DIB Access, or PPMU instruments active). Load a validated test program and restart the debug display to remedy this situation.

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VHFAC Source Debug Display : Debug Display Parameters and Buttons

Debug Display Parameters and Buttons

The VHFAC Source Debug Display contains several sections. Each section has programmable features used to make adjustments to test parameters. The sections are:

- VHFAC Source Schematic
- VHFAC Source Waveform Sourcing Options
- VHFAC Source Trigger Options
- VHFAC Source Pin Setup Information
- VHFAC Source DUT State

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VHFAC Source Debug Display : Debug Display Parameters and Buttons : VHFAC Source Schematic

VHFAC Source Schematic

The VHFAC Source Schematic section displays the current programmed settings of the selected VHFAC Source instrument and allows you to change program parameters. Make connections to the DUT and other instruments by clicking the relay symbols.

The table below identifies the VHFAC Source schematic debug display parameters and buttons.

VHFAC Source Schematic Parameters and Buttons

Parameter/Button	Description
LowPass	This drop-down list contains the size of the filter used for sourcing waveforms. It contains the frequencies that can be selected for the VHFAC Capture low-pass filters: 2.0MHz, 5.5MHz, 10.0MHz, 20.0MHz, 40.0MHz, 80.0MHz, and 160.0MHz
Condition Enable	This checkbox enables condition events.
Condition Connect	This drop-down list contains the destination where the VHFAC Source Instrument passes condition events: • Both • PatGen • Board

MicroCode Control	This drop-down list contains the means by which the microcode control executes: • CPU/Pattern • External Trigger If you select External Trigger, a tab frame appears to show the trigger commands for each source start label.
Coupling/ OutputType	This drop-down list contains the paths of Coupling and OutputType: • ACSingleEnd • DCSingleEnd • ACDifferential • DCDifferential
Amplitude	Sets the amplitude of the signal as it will be seen by the DUT.
CommonMode DC	Sets the CommonMode DC. • Minimum CommonMode DC = -2.0 V • Maximum CommonMode DC = 2.0 V
CommonMode AC	Sets the CommonMode AC. • Minimum CommonMode AC = -2.0 V • Maximum CommonMode AC = 2.0 V

Output Delay	Sets the time delay for the starting point of the waveform. • Maximum Output Delay = 10 us
Load Impedance	Sets the real and imaginary value of load impedance.
AC Runtime Mode	Lists the modes of run-time calibration: • Automatic • Always • None
Signal Frequency	Sets the frequency for leveling. Coupling: AC • Minimum frequency = 5.0 MHz • Maximum frequency = 160.0 MHz Coupling: DC • Minimum frequency = 0.0 MHz • Maximum frequency = 80.0 MHz
PSet	Selects a parameter set (PSet). Selecting a PSet from the list does not apply the settings to hardware. To apply the PSet to hardware, click Apply. Edit brings up the PSet and Signals Editor (PSSE) with the selected PSet loaded. See PSet and Signals Editor in the TDE section.
PPMU	Click this button to open the PPMU debug display for the selected pin. See PPMU Debug Display.

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VHFAC Source Debug Display : Debug Display Parameters and Buttons : VHFAC Source Waveform Sourcing Options

VHFAC Source Waveform Sourcing Options

The block in the upper-left section of the display features input boxes and buttons which allow you to select the wavename for a signal, where the source instrument will receive the start command from, and the sample rate at which the waveform will be sourced. The following parameters and buttons in the VHFAC Source debug display are used to control the source of the signal.

VHFAC Source Waveform Sourcing Parameters and Buttons

Parameter/Button	Description
Signal	Lists the names of the available signals listed in the signal collection. The state label under this drop-down list shows the current state (Stopped/Running) of signal.
View	Starts WaveScope and plots the sourced waveform segment.
Edit	Brings up the PSet and Signals Editor (PSSE) with the selected signal loaded. This control is grayed out if no signals are available. See PSet and Signals Editor in the TDE section.
Load Settings	This button is enabled when a signal name is selected from the Signal field. It loads the settings of the specified Signal. That signal must be defined from VBT.
Start	Manually starts the source of the waveform for debug purposes.
Start option	Lists the options for starting waveform.The following modes can be set:

	• At End • Immediately • Once • Once At End
Stop	Manually stops the sourcing of the waveform when being used for debug purposes.
Stop option	Lists the options for stopping waveform The following modes can be set: • At End • Immediately
SampleRate Enable	This check-box shows the state of SampleRate enable. • Selected: SampleRate Enabled • Cleared: SampleRate Disabled
Sample Rate	The number of samples to source per second. • Minimum sample rate = 3.43 kHz • Maximum sample rate = 400.00 MHz

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VHFAC Source Debug Display : Debug Display Parameters and Buttons : VHFAC Source Trigger Options

VHFAC Source Trigger Options

These options allow you to make waveform sourcing asynchronous to the test system. A change of state on the asynchronous trigger pin triggers a measurement on the pin.

VHFAC Source Trigger Parameters

Parameter/Button	Description
Pin	Specifies the pin whose signal is used to asynchronously trigger the instrument.
ChanType	This is the asynchronous trigger pin's channel type. The trigger channel type should match the asynchronous trigger pin channel type.
Which	A number identifying the trigger line used to asynchronously trigger the instrument.

VHFAC Source Asynchronous Trigger Command Parameters

Parameter/Button	Description
Start Label	Select the start label to view its Trigger commands.
Command	What action will occur.
Label	What Signal it will occur on.

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VHFAC Source Debug Display : Debug Display Parameters and Buttons : VHFAC Source Pin Setup Information

VHFAC Source Pin Setup Information

This information sets the pin and site to be debugged. Once you select a pin, the debug display contains the pin's current setup.

VHFAC Source Pin Setup Display Parameters and Buttons

Parameter/Button	Description
Pin	Lists the available pins from the pin map connected to the Source instrument. Selecting a pin from the list loads parameters for that pin to be debugged.
Site	This drop-down box, located on the debug display tool bar, lists the sites for the selected pin that are available. Selecting a site loads the parameters for the selected pin.
Apply All Sites	This icon is on the debug display tool bar. Clicking this button writes all changes made to every active site. If the button is not depressed, only the current active site is affected by the changes.

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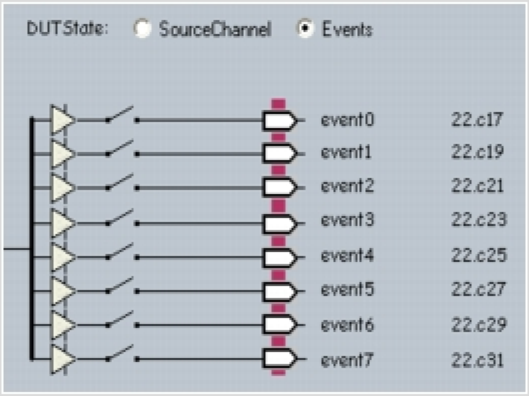
VHFAC Source Debug Display : Debug Display Parameters and Buttons : VHFAC Source DUT State

VHFAC Source DUT State

This section allows you to select between source channel DUT pins and source event signal DUT pins.

VHFAC Source DUT State Display Parameters and Buttons

Parameter/Button	Description
SourceChannel	This selection displays the current state and names of primary or secondary source channel DUT pins.
Events	This selection displays the current state and names of source event signal DUT pins. 

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VHFAC Source Examples

VHFAC Source Examples

This section contains program examples for the VHFAC Source instrument. These IG-XL workbook examples show all IG-XL data, VBT code, necessary for a complete running program.

The following topics are available:

- VHFAC Source Sine Wave (VBT)
- VHFAC Source Sine Wave (Pattern)

For more information, see:

- VHFACSource (VBT Syntax Reference)
- VHFAC Source Programming
- TTN: VHFAC Relay Usage and Life Improvement Guidelines

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VHFAC Source Examples : VHFAC Source Sine Wave (VBT)

VHFAC Source Sine Wave (VBT)

This example uses the VHFAC source instrument to generate a sine wave.

This workbook example includes the following program components:

- DUT Connections
- DataTool Sheets
- VBT Code

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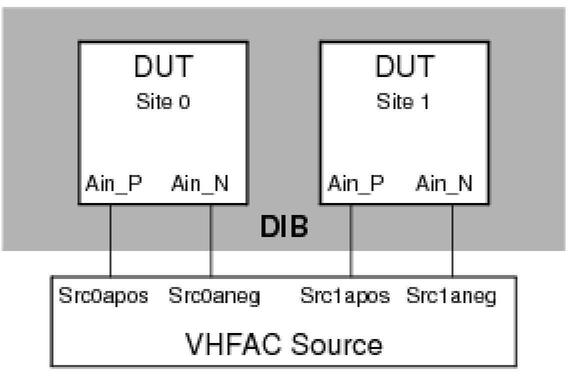
VHFAC Source Examples : VHFAC Source Sine Wave (VBT) : DUT Connections

DUT Connections



VHFAC Source Sine Wave DIB Connections

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VHFAC Source Examples : VHFAC Source Sine Wave (VBT) : DataTool Sheets

DataTool Sheets

This workbook example uses the following DataTool sheets:

- Pin Map Sheet
- Channel Map Sheet
- Test Instances Sheet
- Flow Table Sheet
- Mixed Signal Timing Sheet
- Wave Definitions Sheet

Pin Map Sheet

Pin Map

Group Name	Pin Name	Type
	Ain_P	Analog
	Ain_N	Analog

Channel Map Sheet

Channel Map

DIB ID:

View Mode: Signal

Device Under Test		Tester Channel			
Pin Name	Package Pin	Type	Site 0	Site 1	Co
Ain_P		VHFACSourcePrimaryPos	13.src0apos	13.src1apos	
Ain_N		VHFACSourcePrimaryNeg	13.src0aneg	13.src1aneg	

Test Instances Sheet

Test Instances

TERADYNE

	Test Procedure			DC Specs		AC Specs		Sheet Parameters			
Test Name	Type	Name	Called As	Category	Selector	Category	Selector	Time Sets	Edge Sets	Pin Levels	Mixed Signal Timing
Src15MHz	VBT	VHFACSourceSetupStartStop_VBT									Sine15MHz

The test instance named Src15MHz has the following test-specific parameters as shown in the Instance Editor:

Instance: Src15MHz

Parameters | Timing and Levels | Limits

Name	Value
SrcPin	Ain_P
SrcSigName	Src15MHzSine
SrcAmpVpk	1

Flow Table Sheet

Flow Table

Flow Domain:

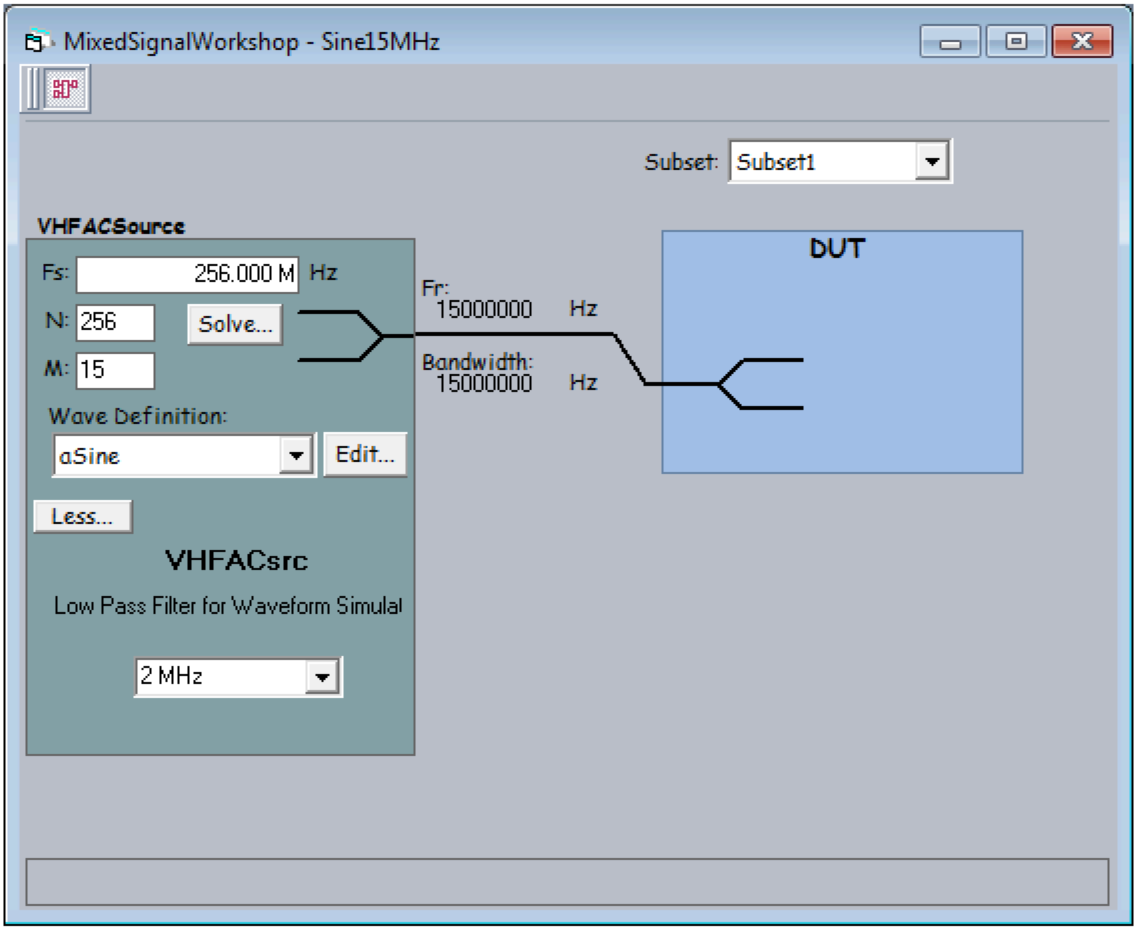
		Gate			Command	
Label	Enable	Job	Part	Env	Opcode	Parameter
					Test	Src15MHz

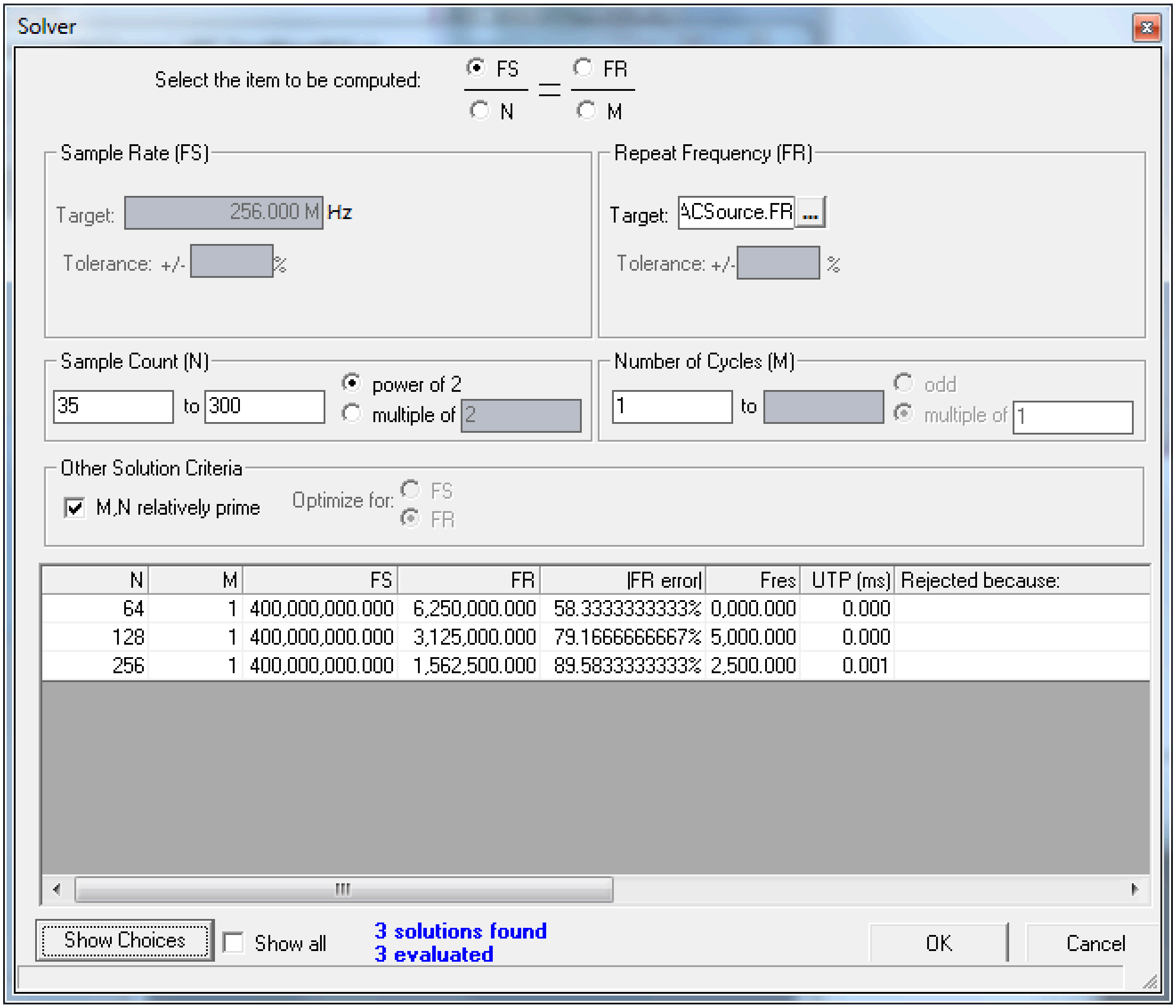
Mixed Signal Timing Sheet

Mixed Signal Timing

TERADYNE

		Resource		Clocking					Instrument	Waveform	
Set Name	Subset	Type	ID	Fs	N	Fr	M	USR	Data	Definition	Filter
Sine15MHz	Subset1	VHFACSource		256000000	256	15000000	15	1		aSine	


MixedSignalWorkshop - Sine15MHz
Subset:
Subset1
VHFACSource
Fs:
256.000 M
Hz
N:
256
Solve...
M:
15
Wave Definition:
aSine
Edit...
Less...
VHFACsrc
Low Pass Filter for Waveform Simulat
2 MHz
Fr:
15000000
Hz
Bandwidth:
15000000
Hz
DUT



Wave Definitions Sheet

Wave Definitions

WaveDefName	WaveDefType	WaveDef Component	Repeat Count	Relative Period	Relative Amplitude	Relative Offset	Primitive Parameters
aSine	Primitive	Sine	1	1.E+00	1.E+00	0	phase=0.0

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VHFAC Source Examples : VHFAC Source Sine Wave (VBT) : VBT Code

VBT Code

```
Public Function VHFACSourceSetupStartStop_VBT(SrcPin As PinList, SrcSigName As String,
SrcAmpVpk As Double) As Long
    On Error GoTo errHandler
    ' Connect VHFAC Source differentially to the DIB
    TheHdw.VHFACSource.Pins(SrcPin).Connect tlVHFACSourceConnectPrimaryPos +
tlVHFACSourceConnectPrimaryNeg, tlVHFACSourceConnectToDUT
    ' Set up the VHFAC Source Signal
    TheHdw.VHFACSource.Pins(SrcPin).Signals.Add SrcSigName
    With TheHdw.VHFACSource.Pins(SrcPin).Signals(SrcSigName)
        .ApplyMixedSignalTiming
        .CalibrationFrequency.Mode = tlCSignalModeUseCalculatedValue
        .LowPass.Bypass = False
        .LowPass.Value = (.SampleRate * .CycleCount / .SampleSize) * 1.25 ' >= 25% more than Fi
        .OutputType = tlVHFACSourceOutputTypeDifferential
        .Coupling = tlVHFACSourceCouplingDC
        .Amplitude = SrcAmpVpk
        .LoadImpedance.Real = 50
        .LoadSettings
    End With
    ' Start the Source
    TheHdw.VHFACSource.Pins(SrcPin).Signals(SrcSigName).Start tlStartImmediately
    ' Let the source run for 5ms
    TheHdw.Wait (0.005)
```

```
    ' Stop the Source
    TheHdw.VHFACSource.Pins(SrcPin).Signals(SrcSigName).Start tlStartImmediately
    Exit Function
errHandler:
    If AbortTest Then Exit Function Else Resume Next
End Function
```

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VHFAC Source Examples : VHFAC Source Sine Wave (Pattern)

VHFAC Source Sine Wave (Pattern)

This example uses the VHFAC source instrument to generate a sine wave.

This workbook example includes the following program components:

- DUT Connections
- DataTool Sheets
- Patterns
- VBT Code

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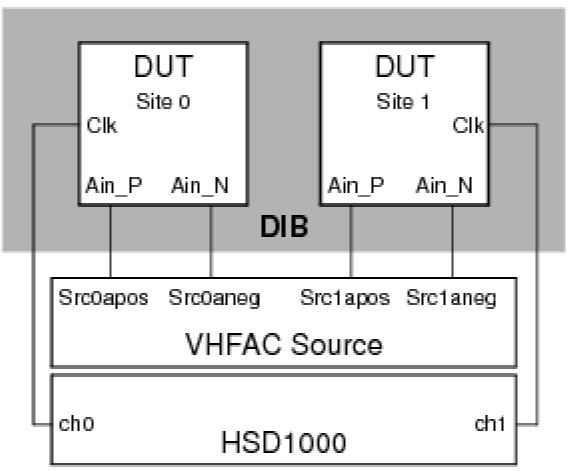
VHFAC Source Examples : VHFAC Source Sine Wave (Pattern) : DUT Connections

DUT Connections



VHFAC Source Sine Wave DIB Connections

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VHFAC Source Examples : VHFAC Source Sine Wave (Pattern) : DataTool Sheets

DataTool Sheets

This workbook example uses the following DataTool sheets:

- Pin Map Sheet
- Channel Map Sheet
- Test Instances Sheet
- Flow Table Sheet
- Mixed Signal Timing Sheet
- Wave Definitions Sheet
- Time Sets (Basic)
- Pin Levels Sheet

Pin Map Sheet

Pin Map

Group Name	Pin Name	Type
	Ain_P	Analog
	Ain_N	Analog
	Clk	Input

Channel Map Sheet

Channel Map

DIB ID: **View Mode:** Signal

Device Under Test		Tester Channel			
Pin Name	Package Pin	Type	Site 0	Site 1	Co
Ain_P		VHFACSourcePrimaryPos	13.src0apos	13.src1apos	
Ain_N		VHFACSourcePrimaryNeg	13.src0aneg	13.src1aneg	
Clk		I/O	16.ch0	16.ch1	

Test Instances Sheet

Test Instances

TERADYNE

	Test Procedure			DC Specs		AC Specs		Sheet Parameters			
Test Name	Type	Name	Called As	Category	Selector	Category	Selector	Time Sets	Edge Sets	Pin Levels	Mixed Signal Timing
SrcAmpSteps15MHz	VBT	VHFACSourceSetupStartStop_POP						TSB		Levels	Sine15MHz

The test instance named SrcAmpSteps15MHz has the following test-specific parameters as shown in the Instance Editor:

Instance: SrcAmpSteps15MHz

Parameters | Timing and Levels | Limits

Name	Value
SrcPin	Ain_P
SrcSigName	Src15MHz1V
SrcAmpVpk	1
patName	.\VHFACSource_POP.P/

Flow Table Sheet

Flow Table

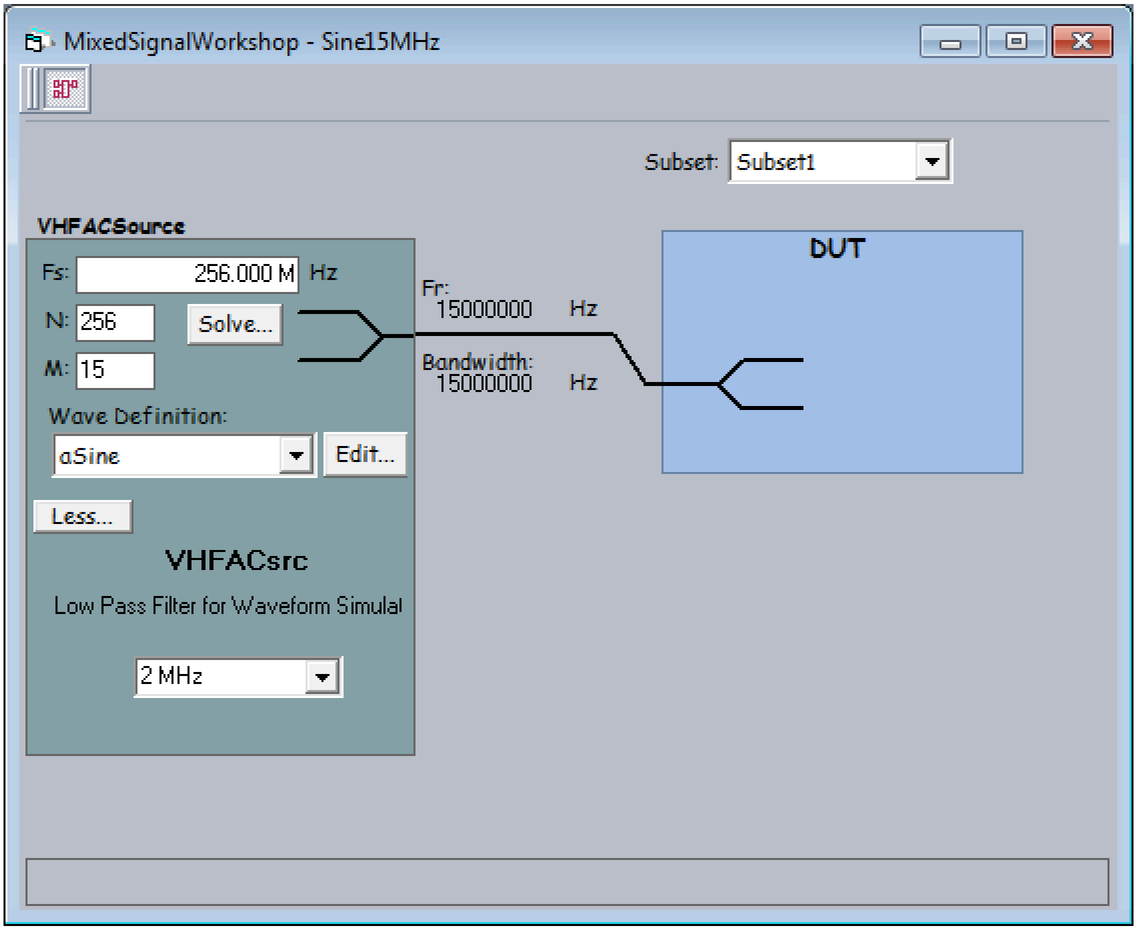
Flow Domain:

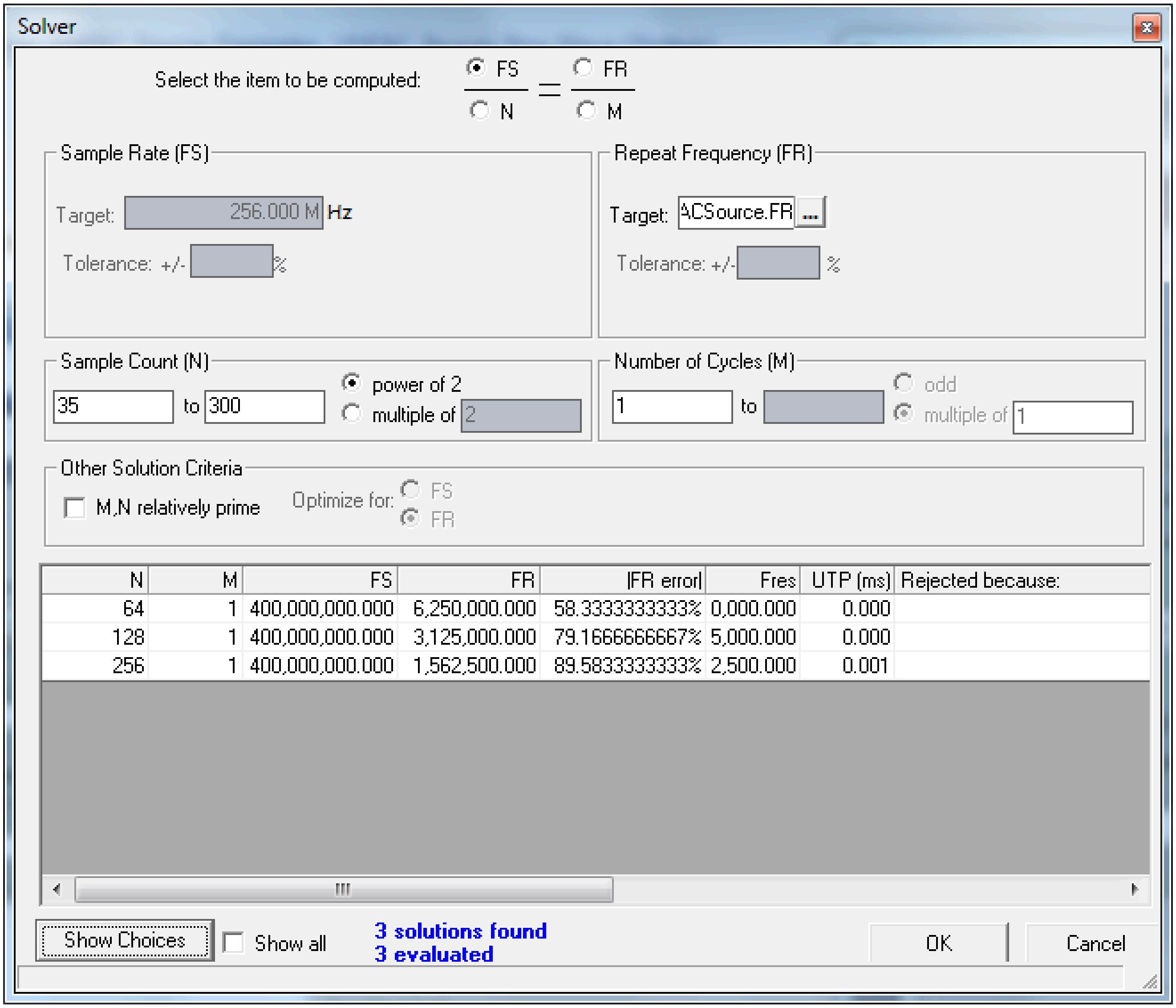
		Gate			Command	
Label	Enable	Job	Part	Env	Opcode	Parameter
					Test	SrcAmpSteps15MHz

Mixed Signal Timing Sheet

Mixed Signal Timing

		Resource		Clocking					Instrument	Waveform	
Set Name	Subset	Type	ID	Fs	N	Fr	M	USR	Data	Definition	Filter
Sine15MHz	Subset1	VHFACSource		256000000	256	15000000	15	1		aSine	





Wave Definitions Sheet

Wave Definitions

WaveDefName	WaveDefType	WaveDef Component	Repeat Count	Relative Period	Relative Amplitude	Relative Offset	Primitive Parameters
aSine	Primitive	Sine	1	1.E+00	1.E+00	0	phase=0.0

Time Sets (Basic)

Time Sets (Basic)

Timing Mode:	Single	Master Timeset Name:	
Time Domain:		Strobe Ref Setup Name:	

Time Set	Cycle Period	Pin/Group Name	Clock Period	Setup	Data Src	Data Fmt	Drive On	Drive Data	Drive Return	Drive Off	Compare Mode	Compare Open	Compare Close	Compare Ref Offset	Edge Resolution Mode
tset_1MHz	1.E-06	Clk			PAT	RH	0	0	500.E-09	1.E-06	Off				Auto

Pin Levels Sheet

Pin Levels

Pin/Group	Seq.	Parameter	Value
Clk		Vil	0
Clk		Vih	3.3
Clk		Vt	0
Clk		Vcl	-1
Clk		Vch	6
Clk		DriverMode	Largeswing-HiZ

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VHFAC Source Examples : VHFAC Source Sine Wave (Pattern) : Patterns

Patterns

TSet	Command	Label	Vector	Ain_P (VHFAC Source)	Clk	Comment
tset_1MHz			0	Start	0	start the Source
tset_1MHz	repeat 5000		1		0	source 1V for 5ms
tset_1MHz			2	PSet 500mV	0	apply PSET for 500mV test
tset_1MHz	repeat 5000		3		0	wait 5ms for PSET to settle
tset_1MHz	repeat 5000		4		0	source 500mV for 5ms
tset_1MHz			5	PSet 250mV	0	apply PSET for 250mV test
tset_1MHz	repeat 5000		6		0	wait 5ms for PSETs to settle
tset_1MHz	repeat 5000		7		0	source 500mV for 5ms
tset_1MHz			8	Stop	0	stop the Source
tset_1MHz	halt		9		0	

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VHFAC Source Examples : VHFAC Source Sine Wave (Pattern) : VBT Code

VBT Code

```
Public Function VHFACSourceSetupStartStop_POP(SrcPin As PinList, SrcSigName As _
    String, SrcAmpVpk As Double, patName As Pattern) As Long

    On Error GoTo errHandler

    ' Apply digital levels and timing to the hardware
    Call TheHdw.Digital.ApplyLevelsTiming(False, True, True, tlPowered)

    ' Connect VHFAC Source differentially to the DIB
TheHdw.VHFACSource.Pins(SrcPin).Connect tlVHFACSourceConnectPrimaryPos + _
    tlVHFACSourceConnectPrimaryNeg, tlVHFACSourceConnectToDUT

' Set up the VHFAC Source Signal
TheHdw.VHFACSource.Pins(SrcPin).Signals.Add SrcSigName

With TheHdw.VHFACSource.Pins(SrcPin).Signals(SrcSigName)
    .ApplyMixedSignalTiming

    .CalibrationFrequency.Mode = tlCSignalModeUseCalculatedValue
    .LowPass.Bypass = False
    .LowPass.Value = (.SampleRate * .CycleCount / .SampleSize) * 1.25 ' >= 25% more than Fi

    .OutputType = tlVHFACSourceOutputTypeDifferential
    .Coupling = tlVHFACSourceCouplingDC
```

```
.Amplitude = SrcAmpVpk
.LoadImpedance.Real = 50

.LoadSettings
 End With

TheHdw.VHFACSource.Pins(SrcPin).Signals.DefaultSignal = SrcSigName

' Set up the VHFAC Source PSets
If TheExec.ExecutionCount = 0 Then ' Set up PSets on first run of program only
   With TheHdw.VHFACSource.Pins(SrcPin).PSets
      .Add "500mV"
      .Add "250mV"
   End With

   With TheHdw.VHFACSource.Pins(SrcPin)
      .PSets("500mV").Amplitude = 0.5
      .PSets("500mV").AmplitudeUnit = tlVHFACSourceAmplitudeUnitVoltPeak
      .PSets("500mV").CalibrationFrequency = .Signals(SrcSigName). _
SampleRate * .Signals(SrcSigName).CycleCount / .Signals(SrcSigName).SampleSize
      .PSets("500mV").CommonModeForAC = _
.Signals(SrcSigName).CommonMode.AC.Value
      .PSets("500mV").CommonModeForDC = .Signals(SrcSigName).CommonMode.DC
      .PSets("500mV").LowPass.Bypass = .Signals(SrcSigName).LowPass.Bypass
      .PSets("500mV").LowPass.Value = .Signals(SrcSigName).LowPass.Value

      .PSets("250mV").Amplitude = 0.25
      .PSets("250mV").AmplitudeUnit = tlVHFACSourceAmplitudeUnitVoltPeak
      .PSets("250mV").CalibrationFrequency = .Signals(SrcSigName). _
SampleRate * .Signals(SrcSigName).CycleCount / .Signals(SrcSigName).SampleSize
      .PSets("250mV").CommonModeForAC = _
.Signals(SrcSigName).CommonMode.AC.Value
      .PSets("250mV").CommonModeForDC = .Signals(SrcSigName).CommonMode.DC
      .PSets("250mV").LowPass.Bypass = .Signals(SrcSigName).LowPass.Bypass
      .PSets("250mV").LowPass.Value = .Signals(SrcSigName).LowPass.Value
   End With
End If
```

```
    ' Start the pattern
    With TheHdw.Patterns(patName)
        .Load
        .Start
    End With

    ' Wait for pattern to halt
    TheHdw.Digital.Patgen.HaltWait

    Exit Function
errHandler:
    If AbortTest Then Exit Function Else Resume Next
End Function
```

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VHFAC Source Programming

VHFAC Source Programming

This section describes programming of a VHFAC Source to source a waveform to a device.

The following topics are available:

- Simulated Configuration Files
- Programming Overview
- VHFAC Source DataTool Programming
- VHFAC Source Pattern Control
- Advanced Programming

For more information, see:

- VHFACSource (VBT Syntax Reference)
- VHFAC Source Examples

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VHFAC Source Programming : Simulated Configuration Files

Simulated Configuration Files

You can use the VHFAC Source offline by forcing IG-XL to map the VHFAC. Do this by creating a SimulatedConfig.txt file and placing the file in any of the following locations:

- Folder containing the IG-XL workbook
- \tester folder: C:\Program Files (x86)Teradyne\IG-XL\<version>\tester
- \bin folder: C:\Program Files (x86)\Teradyne\IG-XL\<version>\bin

The following SimulatedConfig.txt file maps the VHFAC board in slot 13:

```
<Version>
1
</Version>

<TEST_HEAD TH 1>
#slot[.subslot]   Type            idprom (type serial rev company)

13          VHFAC            805-014-00 0000000 0528-A 5445
            VHFAC            939-376-00 0000000 0528-A 5445
            VHFACCapture0    939-244-00 0000000 0450-B 5445
            VHFACTime        939-376-00 0000000 0528-A 5445
            VHFACSource5     939-247-10 0000000 0450-B 5445
            VHFACSource4     939-247-10 0000000 0450-B 5445
            VHFACCapture1    939-244-00 0000000 0450-B 5445
```

24.0 SupportBoard 974-220-12 0000000 0000-A 0000
SupportBoard 956-354-00 0000000 0000-A 0000

62.0 DistributionBoard 956-355-00 0000000 0000-A 0000
DistributionBoard 956-355-00 0000000 0000-A 0000

</TEST_HEAD TH 1>

To use the VHFAC Source offline, the SimulatedConfig.txt file must include every board needed in the IG-XL workbook. The SimulatedConfig.txt file represents the state of the offline tester. To use the VHFAC Source online, you need not edit any files. IG-XL automatically maps all boards in the tester.

For more information, see Tester Configuration.

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VHFAC Source Programming : Programming Overview

Programming Overview

To develop a test program that uses the VHFAC Source instrument, follow this basic procedure:

1. Create a pin map with pins of the DUT (Programming the Pin Map)
2. Create a channel map that allocates the tester resources to the DUT pins (Programming the Channel Map).
3. Create a mixed signal timing set (Creating a Wave Definition and Mixed Signal Timing Set).
4. Create a test flow, the order in which the test will execute, in the Flow Table sheet.
   - Create a test function using VBT code in the Visual Basic editor.
   - Instantiate the VBT test and set the test values in the Instance Editor.
5. Validate and run the test program.

VBT syntax is described in the VHFACSource section of the VBT Syntax Reference.

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VHFAC Source Programming : VHFAC Source DataTool Programming

VHFAC Source DataTool Programming

This section describes how to program several DataTool worksheets to handle basic VHFAC Source parameters. It includes the following topics:

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VHFAC Source Programming : VHFAC Source DataTool Programming : Programming the Pin Map

Programming the Pin Map

A pin map describes the device pins by assigning a pin type to each device pin name. For the VHFAC Source, the connections that you must enter in the pin map are analog channel connections to the DUT.

The valid Type selection for the VHFAC Source instrument is:

Analog	This type is used for the VHFAC Source instrument.

☑ **Note:**

If DIB Access or Multiple Resources are used for a device pin and the VHFAC Source is not the primary resource, you can select a different type for the device pin.

For more information, see Pin Map Sheet in the DataTool section.

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VHFAC Source Programming : VHFAC Source DataTool Programming : Programming the Channel Map

Programming the Channel Map

A Channel Map sheet assigns individual device pin names to test system channels and power supplies. For the VHFAC Source instrument, you must select a channel type from the Type column menu for each pin or pin group.

Valid VHFAC Source channel types are:

VHFACSourcePrimaryNeg	VHFAC Source Primary Negative Pin
VHFACSourcePrimaryPos	VHFAC Source Primary Positive Pin
VHFACSourceSecondaryNeg	VHFAC Source Secondary Negative Pin
VHFACSourceSecondaryPos	VHFAC Source Secondary Positive Pin
VHFACSourceEvent[0–7]	VHFAC Event Lines (8 event lines)

Type the tester channel information value for each site to connect a DUT pin to the instrument. Note that in the table, x indicates the DIB slot number of the VHFAC Source.

VHFAC Source Channel Resources

VHFAC Source Pogo Pin	DIB Channel	Signal Name
VHFAC Source 0 Primary Pos	x.a3	x.src0apos
VHFAC Source 0 Primary Neg	x.a7	x.src0aneg
VHFAC Source 0 Secondary Pos	x.c3	x.src0bpos

VHFAC Source 0 Secondary Neg	x.c7	x.src0bneg
VHFAC Source 1 Primary Pos	x.b2	x.src1apos
VHFAC Source 1 Primary Neg	x.b6	x.src1aneg
VHFAC Source 1 Secondary Pos	x.d2	x.src1bpos
VHFAC Source 1 Secondary neg	x.d6	x.src1bneg
Event 0 Lines	x.c17	x.event0_0
	x.c19	x.event0_1
	x.c21	x.event0_2
	x.c23	x.event0_3
	x.c25	x.event0_4
	x.c27	x.event0_5
	x.c29	x.event0_6
	x.c31	x.event0_7
Event 1 Lines	x.d16	x.event1_0
	x.d18	x.event1_1
	x.d20	x.event1_2
	x.d22	x.event1_3
	x.d24	x.event1_4
	x.d26	x.event1_5
	x.d28	x.event1_6
	x.d30	x.event1_7

For more information, see Channel Map Sheet in the DataTool section.

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VHFAC Source Programming : VHFAC Source DataTool Programming : Creating a Wave Definition and Mixed Signal Timing Set

Creating a Wave Definition and Mixed Signal Timing Set

The major function of the VHFAC Source instrument is to generate and deliver analog waveforms of a specific shape, frequency, and amplitude to the DUT. These waveforms start as wave definitions, which define the basic shape of the waveform. Then a timing solution is added to the waveform definition to create a timing set for the test. The VHFAC Source instrument uses the timing set to create a reusable signal to which various frequency and amplitudes can be applied.

The following topics provide step-by-step instructions for creating a wave definition and a timing context:

- Setting Up the Wave Definitions Sheet
- Creating a Timing Set Using the Mixed Signal Workshop

Setting Up the Wave Definitions Sheet

The Wave Definitions sheet contains the wave shape information of waveform definitions that were created using the WaveDesigner. This sheet stores the wave definition information and the test program accesses the information when the test is run.

To insert a Wave Definitions sheet:

1. Select Insert>Insert IG-XL Worksheet in the Sheet group on the IG-XL Main tab.
2. Select Wave Definitions from the Sheet type drop-down list in the Insert IG-XL Worksheet dialog box.

3.	Type a name for the sheet in the Sheet name box (optional). The default name is WaveDef.

4.	Click OK.

By default, an initial wave definition called aSine is added to the sheet.

Creating a Timing Set Using the Mixed Signal Workshop

The Mixed Signal Workshop (MSW) tool helps you to build reusable mixed signal timing solutions that meet coherence requirements.

The WaveDesigner tool creates wave definitions, which are independent of timing and instrument specific information. The MSW uses these wave definitions and then applies the instrument specific timing information necessary to generate samples.

The output of the MSW is a mixed signal timing set which contains the waveform's structural characteristics and the details of the clocking solution that you created. Timing sets are stored on the Mixed Signal Timing sheet in the IG-XL workbook and become a reusable timing solution for future tests.

To create the required timing set:

1.	Insert and set up a Mixed Signal Timing sheet in the IG-XL workbook.

a.	Select Insert>Insert IG-XL Worksheet in the Sheet group on the IG-XL Main tab.

b.	Select Mixed Signal Timing from the Sheet type drop-down list in the Insert IG-XL Worksheet dialog box.

c.	Type a name for the sheet in the Sheet name box (optional). Default name is Mixed Signal Timing.

d.	Click OK.

e.	Type a timing context name in the Set Name column and click Enter.

f.	Right-click the timing set name and select Edit from the context menu to start the Mixed Signal Workshop (MSW) tool in TDE.

The MSW display appears and displays the DUT and capture resources blocks.

2. Add a VHFAC Source resource to the MSW display:

a. Select MSW>Add Source.

b. Select VHFAC Source from the Select Resource dialog box.
Leave the Resource ID box empty, unless this timing context is using additional VHFAC Source instruments. If that is the case, assign the current source instrument a unique ID number to distinguish it from the others.

c. Click the OK in the Select Resource dialog box.
The VHFAC Source resource block is added to the MSW display.

☑ **Note:**
MSW launches a blank set with a default Capture instrument resource. If you do not require this resource, select the block, then right-click. Choose Delete instrument from the menu.

3. Select a wave definition from the WaveDefinition menu on the VHFAC Source resource block.

4. Specify the Source timing. Typically, the objective of programming the VHFAC Source instrument is to find a sample rate (Fs) and sample size (N) that achieve the desired source frequency.

a. Click the Solve button in the Source resource block to start the Solver tool.

b. Select Fs to solve for the sample rate (Fs).

c. In the Repeat Frequency frame, type in the value for the desired repeat frequency (FR) or, click the ellipsis button (...) to choose from a list of desired capture frequencies. Choosing from the list of desired capture frequencies is often preferred because it allows the VHFAC Source to be programmed based on the capture's sampling parameter constraints. Note that for a sine wave, the repeat frequency is the same as the fundamental frequency (Fi).

d. Constrain the sample count to return a useful number of solutions by modifying the starting and stopping values in the Sample Count frame. In certain cases, the MSW default search parameters for sample size do not yield a solution, and you must increase the upper sample size limit.

e. Enter the number of cycles. Typically, the desired number of cycles (M) for the source waveform is 1. This is because this results in the minimum number of source samples and because the source instrument will repeat the segment indefinitely. The MSW Solver's tool default is set to 1.

f. Click the Show Choices button, select a solution from the table, and click OK.
Typically, the solution with a minimum N and/or minimum FR error is desired. However, the best solution can depend on the specifics of your test's needs. Note that if no solutions were found, you must modify the inputs to the Solver. Typically, this is sample size range that was used. Additionally, you can select Show All to see why some of the solutions were not valid.

5. Save the timing set to the Mixed Signal Timing sheet by clicking the Save to Excel button in the toolbar. Note that this does not save the workbook, only the timing set to the MST sheet.

For more information, see Mixed Signal Workshop.

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VHFAC Source Programming : VHFAC Source Pattern Control

VHFAC Source Pattern Control

The VHFAC Source instrument can be controlled using patterns and microcode, allowing changes to instrument states and parameters during a program. Repeatable timing can be obtained by programming an instrument with microcodes and parameter sets (PSets).

A pattern is created independently of the test program by using the PatternTool or a text editor. Patterns are composed of a series of numbered digital vectors. These digital vectors define the data for each channel listed in the pattern, as well as defining the pattern microcode used for controlling pattern execution.

Microcode	A microcode controls a single function, such as starting or stopping the sourcing of a waveform.
PSets	A VHFAC Source PSet contains parameters such as amplitude and low-pass filter values. When an instrument receives a PSet from the pattern, it goes to the programmed state represented by that PSet.

The following topics are available:

- VHFAC Source Pattern Microcodes
- PSets

Creating a Pattern for Use in a Test Instance

You can create a VHFAC Source pattern by using the PatternTool within TDE or by writing the pattern as an ASCII text file. For small patterns, use the PatternTool. Note that you must first have a valid pin map. Then:

1.	Start the PatternTool by selecting TDE>Tools>Pattern Tool from the menu bar in the TDE window.
2.	Open a window for a new pattern by clicking File>New.
3.	Select single or dual mode and other options on the Options worksheet.
4.	Read in the pin map by clicking File>Read Pinmap.
5.	Create a Pin List on the Pin List worksheet.
6.	Create Pin Setups and define instruments on the Pin Setups and Instruments worksheets if required.
7.	Import Tsets and Labels.
8.	Add Labels, Opcodes, and Vectors on the Vectors worksheet.

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VHFAC Source Programming : VHFAC Source Pattern Control : VHFAC Source Pattern Microcodes

VHFAC Source Pattern Microcodes

A series of extended microcode commands are available for use inside a pattern. Patterns allow you to control the precise timing of when a waveform is sourced. In a case where you want to establish repeatable timing, or you need to synchronize with other instruments (such as the HSD) in the tester, you will want to use a pattern. However, if you want to start generating a waveform but are not concerned with precise timing, then you do not need a pattern.

The following extended microcodes are responsible for issuing commands to VHFAC Source instrument from the pattern:

VHFAC Source Pattern Microcodes

Microcode	Functional Description
Disable_Inst_Cond	Enables the VHFAC Capture to drive the condition bit.
Enable_Inst_Cond	Disables the VHFAC Capture from driving the condition bit.
PSet <name>	Executes the parameter set (PSet) which changes the instrument’s setup (amplitude, low pass filter, and so on). The PSet microcode requires a single argument, which is the name of the PSet as defined in the VBT interpose function.
Resync	Synchronizes the instrument clocks with the pattern clocks to ensure test repeatability after this point.
Start	The instrument immediately (on the next available logical analog clock) starts sourcing the named waveform in SMEM and repeats it continuously until a Stop microcode is executed, the instrument is reset, or another waveform is started.

	The Start microcode requires a single argument, which is the name of the waveform to be sourced. If the default signal name is specified, the Start microcode requires no signal argument.
StartE	At the end of the currently sourced waveform, the instrument sources the named waveform in SMEM and repeats it continuously until a Stop microcode is executed, the instrument is reset, or another waveform is started. The StartE microcode requires a single argument, which is the name of the waveform to be sourced. If the default signal name is specified, the StartE microcode requires no signal argument.
Start1	The instrument immediately (on the next available logical analog clock) starts sourcing the named waveform in SMEM and sources it exactly once. If the default signal name is specified, the Start1 microcode requires no signal argument.
Start1E	At the end of the currently sourced waveform, the instrument sources the named waveform in SMEM and sources it exactly once. The StartE microcode requires a single argument, which is the name of the waveform to be sourced. If the default signal name is specified, the Start1E microcode requires no signal argument.
Stop	The instrument stops sourcing the currently sourced waveform immediately. Stop does not affect a waveform that is running once.
StopE	The instrument stops sourcing the currently sourced waveform at the end of the waveform.

VHFAC Source Microcode Rules

When using source microcodes, follow the rules below:

- When issuing multiple VHFAC Source “Start” microcode commands in a pattern (this includes Start, Start1, StartE, Start1E), the spacing between successive “Start” commands must obey the following rules. Failure to follow these rules will result in a “Microcode Tossing Error”:
  - For source sample rates less than 100 MHz, the time between “Start” microcode commands must be greater than 1/source sample rate.
  - For source sample rates between 100 MHz and 200MHz, the time between “Start” microcode commands must be greater than 2/source sample rate.

- For source sample rates greater than 200MHz, the time between “Start” microcode commands must be greater than 4/source sample rate.

- When issuing a “Start” microcode following a StopE microcode, the time between the microcodes must be greater than the length of the signal being stopped by the StopE. Failure to follow this rule will result in a “Microcode Overwritten Error.”

- When issuing a Resync microcode, the time between the Resync and any other microcode must be greater than 6.5 ms to allow for the source channel clock to come into phase alignment with the high speed digital T0 clock.

- When issuing a PSet microcode, the time between the PSet and any other microcode must be greater than 5 ms to allow for hardware settling.

- For Enable_Inst_Cond and Disable_Inst_Cond microcode commands, there is no limitation for the spacing between successive microcodes.

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VHFAC Source Programming : VHFAC Source Pattern Control : PSets

PSets

A parameter set (PSet) is a set of values that modifies instrument hardware setting using a pattern microcode. PSets allow the test program to change instrument setups without stopping and later resuming pattern execution. PSets improve repeatability because they operate independent of the host computer. This also increases throughput by reducing the amount of handshaking required between the pattern and the host computer. Although PSets are a hardware capability that changes the setup of an instrument during a pattern burst, they are defined in the VBT language and invoked using PSet microcode.

PSet properties and methods for the VHFAC Source instrument are described in the following sections. Note that PSets do not support incremental programming and that only a subset of instrument parameters can be controlled by PSets.

For more information on VHFAC Source VBT syntax, see VHFACSource in the VBT Syntax Reference.

Adding a VHFAC Source PSet to a Test Program

To add VHFAC Source PSet function to a test program do the following:

1.	Use the Add statement to add a specific name to the instrument instance. For example, to add a new PSet “a” to all instrument instances behind pin “Ain”, the code would look like this:

```
TheHdw.VHFACSource.Pins("Ain").PSets.Add ("a")
```

2.	Use the Set statement to set all PSet parameters.

To continue the example, set all parameters of PSet “a” (Amplitude = 2.2, Amp Unit = Vpk, Common Mode DC Offset = 1.1, Common Mode AC offset = 0, Calibration Frequency = 4MHz, Filter Bypass = False, Filter Value = 10MHz), using the following code:

```
Call TheHdw.VHFACSource.Pins("Ain").PSets("a").Set(2.2, _
        tlVHFACSourceAmplitudeUnitVoltPeak, _
        1.1, 0, 4000000#, False, 10000000#)
```

3.	Use the Apply statement to apply the PSet to the indicated pins. In the example, you would apply PSet “a” from VBT as follows:

```
TheHdw.VHFACSource.Pins("Ain").PSets("a").Apply
```

☑ **Note:**

On applying a PSet, a hardware settling time of > 5ms is required. This settle time is not managed by the hardware and software. It is your responsibility.

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VHFAC Source Programming : Advanced Programming

Advanced Programming

Several advanced features for the VHFAC Source instrument are used for specific purposes. These include:

- VHFAC Source Compound Segments
- VHFAC Source Event Lines
- VHFAC Source Condition Bit
- VHFAC Source GainHardware Parameter

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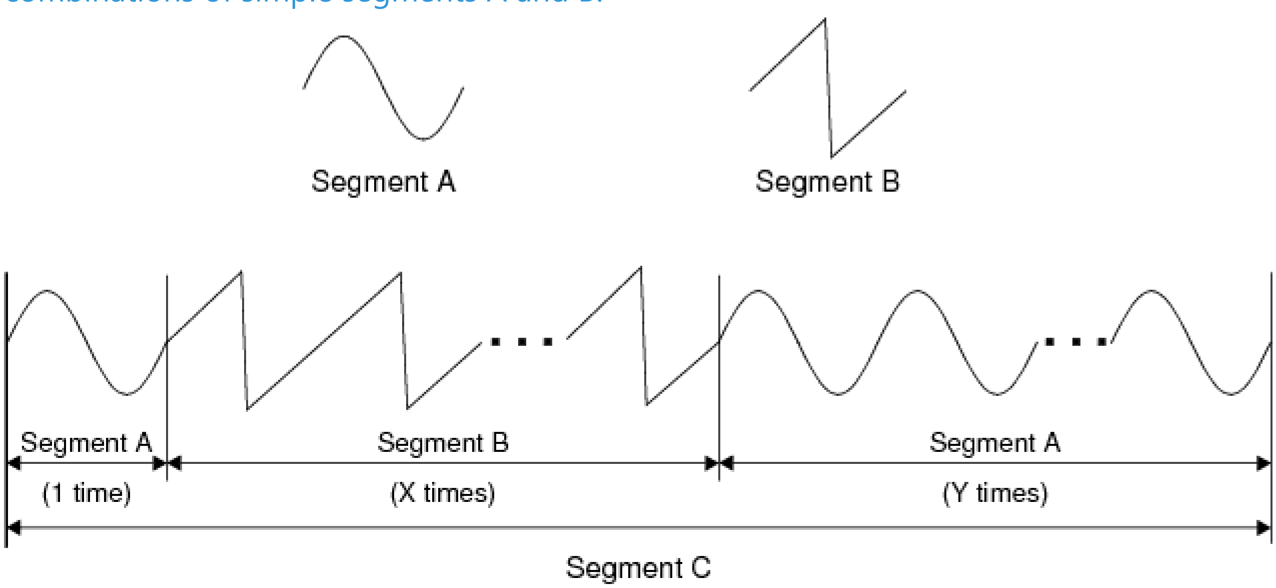
VHFAC Source Programming : Advanced Programming : VHFAC Source Compound Segments

VHFAC Source Compound Segments

What is a Compound Segment?

Compound segments are larger, more complex segments made up of smaller, simple segments. With compound segments you can create large segments while conserving sample memory. For example, in the figure below the complex segment C is a compound segment made up of many combinations of simple segments A and B.



Defining Compound Segments

If the total sample size of a compound segment is <=1048576, then only the total sample size of the compound segment must follow the modulo-N rules dictated by sample rate. If the total sample size of a compound segment is greater than 1048576, then each of the simple segments that make up the compound segment must follow the modulo restrictions dictated by sample rate. For more information, see VHFAC Source Supported Modes Summary and the waveform segment modulus specification in VHFAC Specifications.

Programming Features

The following are programming features of compound segments:

- 4096 sequencer memory size: memory used to store microcodes for generating compound segments at run time.
- Eight levels of nesting allowed (that is, levels of compound segments being used to make larger, more complex compound segments).
- IG-XL requires the use of the “Instrwave” object in VBT, rather than “primitives” in WaveDesigner, to create “wave definitions” that are true compound segments. You cannot create true compound segments (those which conserve SMEM) using WaveDesigner.

For more information, see InstrWave in the VBT Syntax Reference.

VBT Programming Example

```
Public Function CompoundSegmentExample() As Long
    Const aSize1 = 20
    Const aSize2 = 40
    Dim dArray1(aSize1 - 1) As Double
    Dim dArray2(aSize2 - 1) As Double
    Dim WaveObj1 As New InstrWave
    Dim WaveObj2 As New InstrWave
    Dim WaveObj3 As New InstrWave
    Dim i As Long
    Const Pi As Double = 3.14159265358979
    If TheExec.WaveDefinitions.Exists("VBTCompDef") = False Then
        ' Create Sine
        For i = 0 To aSize1 - 1
            dArray1(i) = Sin(2 * Pi * i / aSize1)
        Next i
        ' Create Pulse
        For i = 0 To aSize2 - 1
            If i < aSize2 / 2 Then
                dArray2(i) = 1#
            Else
                dArray2(i) = 0
            End If
        Next i
```

```
        WaveObj1.Add dArray1, 2
        WaveObj2.Add dArray2
        WaveObj3.Add WaveObj1
        WaveObj3.Add WaveObj2
        TheExec.WaveDefinitions.CreateWaveDefinition "VBTCompDef", WaveObj3, False
    End If
    TheHdw.VHFACSource.Pins("SrcPos").Signals.Add "Sig1"
    With TheHdw.VHFACSource.Pins("SrcPos").Signals("Sig1")
        .SampleRate.Value = 400000000#
        .WaveDefinitionName = "VBTCompDef"
    End With
End Function
```

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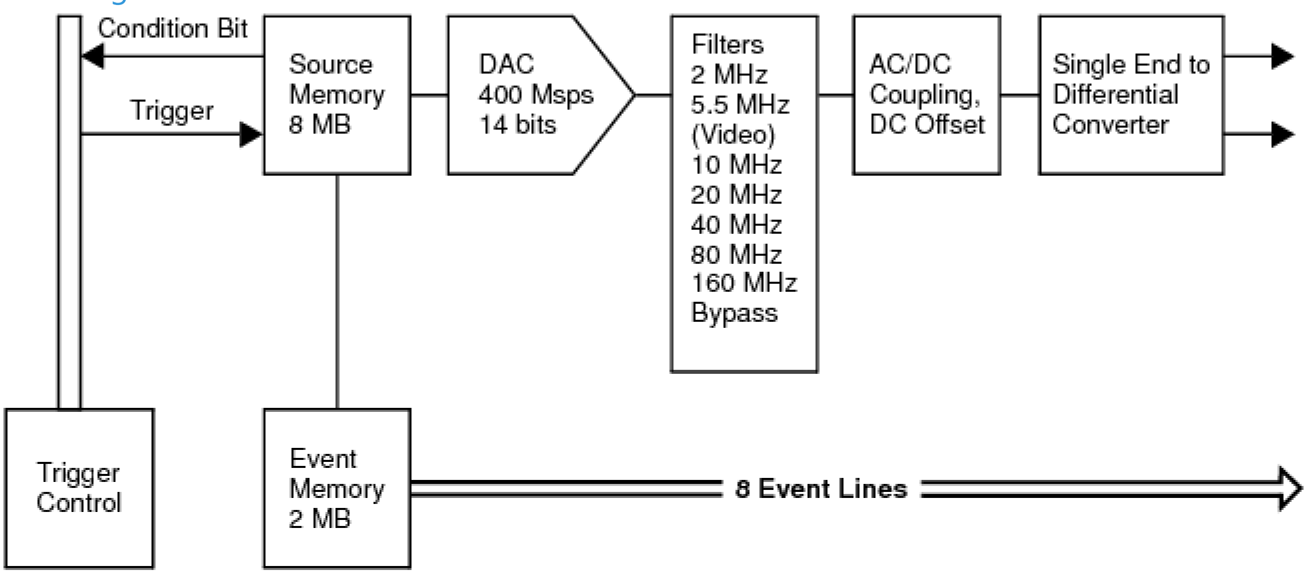
VHFAC Source Programming : Advanced Programming : VHFAC Source Event Lines

VHFAC Source Event Lines

The VHFAC Source can output event signals, which are digital signals synchronized with an analog waveform in the source memory. Event data is output to a DUT through eight Pogo pins as shown in the figure.



Source Event Lines

Event lines are primarily intended for use in video applications, where multiple digital signals synchronized with particular source waveforms are required for precise DUT setup and control. Other applications requiring synchronization signals on the DIB may also benefit from this feature. See VHFAC Source DUT State for more information on the event lines.

VBT Programming Example

Event lines can be programmed using VBT. The following code illustrates this.

VBT Code

```
If TheExec.WaveDefinitions.Exists("SineWithEvents") = False Then
    ' Create Sine
```

```
        For i = 0 To aSize - 1
            dArray(i) = Sin((2 * Pi * Fi / Fs) * i)
        Next i

        WaveObj.Add dArray

        ' Define each Event signal as part of the segment
        With TheHdw.VHFACSource.EventData(WaveObj)
            .Add tlVHFACSourceEvent0, 1, 64
            .Add tlVHFACSourceEvent1, 69, 132
            .Add tlVHFACSourceEvent2, 137, 200
            .Add tlVHFACSourceEvent3, 205, 268
            .Add tlVHFACSourceEvent4, 273, 336
            .Add tlVHFACSourceEvent5, 341, 404
            .Add tlVHFACSourceEvent6, 409, 472
            .Add tlVHFACSourceEvent7, 477, 540
        End With

        TheExec.WaveDefinitions.CreateWaveDefinition "SineWithEvents", WaveObj, False
    End If
```

Comments

In this VBT code example, this line creates a Sine wave to be used to create a wave definition:

dArray(i) = Sin((2 * Pi * Fi / Fs) * i)

This code adds an event signal to the wave object defined by the dArray:

```
With TheHdw.VHFACSource.EventData(WaveObj)
    .Add ...
    .
    .
    .
End With
```

The event lines themselves are indicated by including their number along with their start and stop value. For instance tlVHFACSourceEvent0 indicates event line 0. The first number 1 is the sample value to start the event data, and the second number is the sample value to stop the event data:

.Add tlVHFACSourceEvent0, 1, 64

Note that the start number (in this case 1) must always be (modulo 4) + 1, and the stop number (in this case 64) must always be (modulo 4) or the last sample.

Finally the WaveDefinitions command is used to create the wave definition containing the samples and event data:

```
TheExec.WaveDefinitions.CreateWaveDefinition "SineWithEvents", WaveObj, False
```

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VHFAC Source Programming : Advanced Programming : VHFAC Source Condition Bit

VHFAC Source Condition Bit

In addition to event lines, the VHFAC source instrument can generate a condition bit that synchronizes with an analog waveform in the source memory. While this functionality is similar to event lines, the difference is that the event line data is fed directly to the DUT through Pogo pins, whereas the condition bit is distributed to other instruments on the VHFAC board through the Trigger Control instrument or the condition bit can be sent directly to the PatGen.

The source condition bit is a digital signal programmed to start and stop (go high and low) with specific samples of the source waveform.

The condition bit is routed with VBT commands. To route to the board for use with the Trigger Control (that is, to distribute the condition signal to other VHFAC instruments or to DIB/DUT resources) use the following code:

TheHdw.VHFACSource.Pins(SrcPin).Condition.Connect tlVHFACSourceConditionToBoard

To route to the PatGen, to be used for conditional looping/branching within a pattern, use the following code:

TheHdw.VHFACSource.Pins(SrcPin).Condition.Connect tlVHFACSourceConditionToPatgen

To route to both the VHFAC board and PatGen simultaneously, use the following code:

TheHdw.VHFACSource.Pins(SrcPin).Condition.Connect tlVHFACSourceConditionToBoth

VBT Programming Example

The following code shows how to program the source condition bits:

```
If TheExec.WaveDefinitions.Exists("SineWithCondition") = False Then
    ' Create Sine
    For i = 0 To aSize - 1
        dArray(i) = Sin((2 * Pi * Fi / Fs) * i)
    Next i
```

```
    WaveObj.Add dArray

    ' Define the Condition signal as part of the segment
    TheHdw.VHFACSource.ConditionData(WaveObj).Add 1, 4
    TheExec.WaveDefinitions.CreateWaveDefinition "SineWithCondition", _
        WaveObj, False
End If
' Connect and Enable the Condition output of the Source
With TheHdw.VHFACSource.Pins(SrcPin).Condition
    .Connect tlVHFACSourceConditionToBoth
    .Enable = True
End With
```

Comments

In this VBT code example, this line creates a Sine wave to be used to create a wave definition:

dArray(i) = Sin((2 * Pi * Fi / Fs) * i)

This code adds the condition data to the waveform sample:

TheHdw.VHFACSource.ConditionData(WaveObj).Add 1, 4

The first number after the .Add keyword indicates the sample number to start the condition data. This number must always be (modulo 4) + 1. The second number indicates the sample number to stop the condition data. This number must be (modulo 4) or last sample.

This code creates the wave definition containing the samples and condition data for the wave object defined by dArray:

TheExec.WaveDefinitions.CreateWaveDefinition "SineWithCondition", WaveObj, False

The code between the With and End with keywords determines where to send the condition data (here to both the PatGen and the board for use by other instruments) and enables the condition output:

```
With TheHdw.VHFACSource.Pins(SrcPin).Condition
    .Connect tlVHFACSourceConditionToBoth
    .Enable = True
End With
```

The condition output can also be enabled from the pattern.

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VHFAC Source Programming : Advanced Programming : VHFAC Source GainHardware Parameter

VHFAC Source GainHardware Parameter

Source GainHardware Parameter

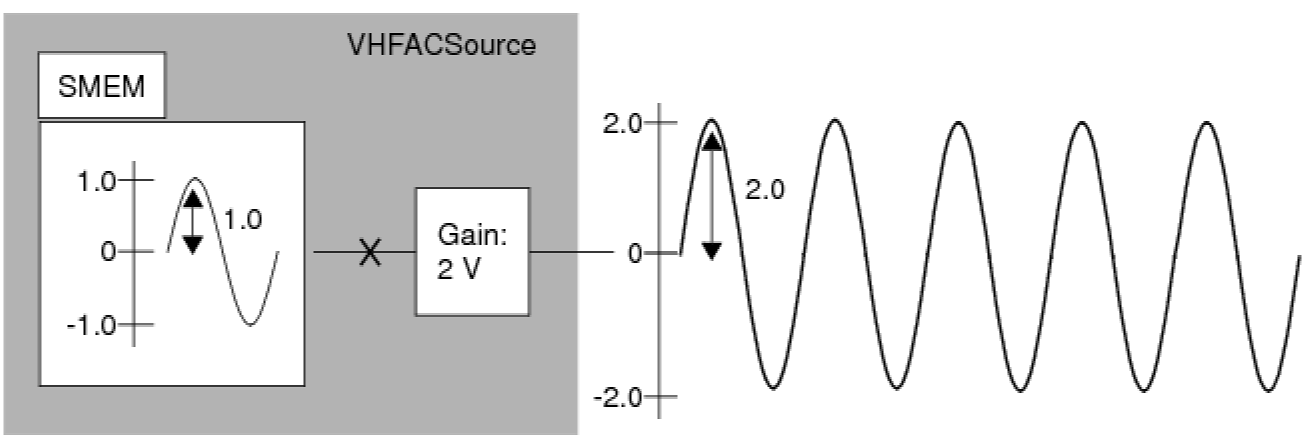
The VHFAC Source VBT Signals language includes the GainHardware parameter. This parameter allows for programmatic access to the Amplitude Control hardware of the VHFAC Source Board.

In most cases, you need not program this parameter. You can leave it in its default state. In some cases, however, you may need to access the Amplitude Control.

GainHardware Example Use Case

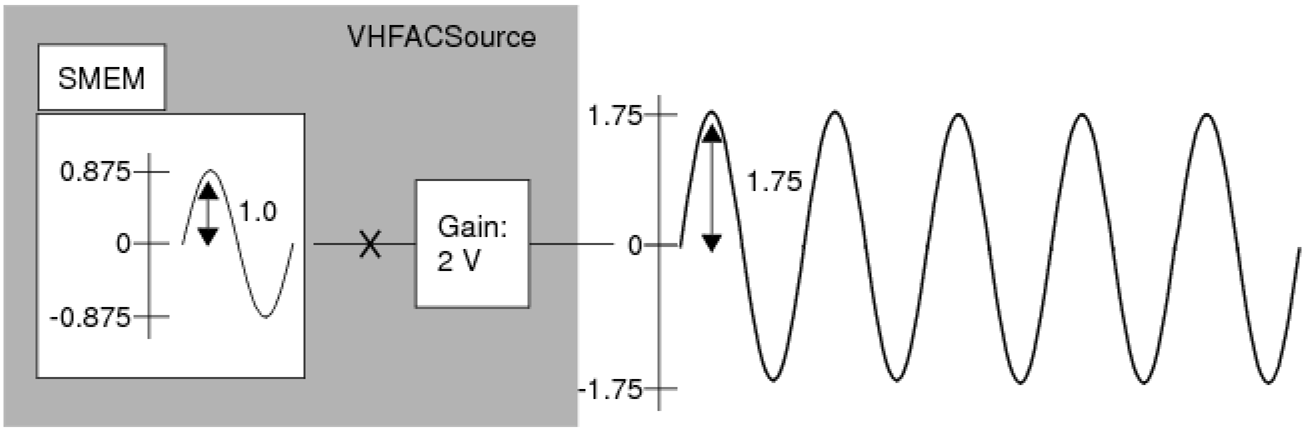
Suppose you want segments that are scaled in SMEM so that signals of different amplitudes can be sourced sequentially, without any change to the hardware. Specifically:

- First, source "myWave1" with a 2 V peak amplitude, 0 V offset. In this case, the instrument will use the full dynamic range of the DAC by including normalized samples in SMEM. The desired signal amplitude of 2 V peak will be realized by applying the instrument hardware gain of 2 V.



Sourcing myWave1 with a 2 V Peak Amplitude

- Second, source “myWave2” with a 1.75 V peak amplitude, 0 V offset. When making the transition between the first and second waveform, do not change to the VHFAC Source amplitude control hardware. In other words, realize the change in amplitude by scaling the samples in SMEM. See the following figure and note the instrument hardware gain remains at 2V while the samples in SMEM are no longer normalized.



Sourcing myWave2 with a 1.75 V Peak Amplitude

Programming the GainHardware Parameter

You can program the GainHardware parameter in the following ways:

- Set GainHardware.Mode to “Use Calculated Value.”

With the “Use Calculated Value” setting, samples are always normalized to 1 and the gain hardware is selected by IG-XL to achieve the desired Signal/WaveDefinition amplitude. This is the default behavior of IG-XL and the default value of the GainHardware parameter.

VBT example:

```
TheHdw.VHFACSource.Pins("Ain").Signals("Sig1").GainHardware.Mode = _
                    tlCSignalModeUseCalculatedValue
```

- Set GainHardware.Mode to “Use Loaded Value.”

This instructs the signal to examine the currently programmed value for Pins(pin).Amplitude and to realize the desired Signal/WaveDefinition Amplitude with no change to hardware, thereby putting the amplitude in the samples.

VBT example:

```
TheHdw.VHFACSource.Pins("Ain").Signals("Sig1").GainHardware.Mode = _
                    tlCSignalModeUseLoadedValue
```

- Program the GainHardware mode to an explicit, numerical value.

IG-XL determines the correct sample/segment/SMEM amplitude and offset depending on the constraints given to the hardware by the Signal.

VBT example:

TheHdw.VHFACSource.Pins("Ain").Signals("Sig1").GainHardware.Value = 2.0

In summary, the options are:

CalculatedValue	No constraints. IG-XL manages scaling of SMEM samples and hardware settings.
UseLoadedValue	The constraint is to not change hardware.
Programmed Value	The constraint is to use the programmed value.

Code Example for the Use Case

To accomplish the tasks for the use case described above, use the following code:

TheHdw.VHFACSource.Pins(SrcPin).Signals.Add ("myWave1, myWave2")

```
With TheHdw.VHFACSource.Pins(SrcPin).Signals("myWave1")
    .ApplyMixedSignalTiming

    .Amplitude = 2#
    .GainHardware.Mode = tlCSignalModeUseCalculatedValue
End With

With TheHdw.VHFACSource.Pins(SrcPin).Signals("myWave2")
    .ApplyMixedSignalTiming

    .Amplitude = 1.75
    .GainHardware.Mode = tlCSignalModeUseLoadedValue
End With

TheHdw.VHFACSource.Pins(SrcPin).Signals("myWave1").LoadSettings

TheHdw.VHFACSource.Pins(SrcPin).Signals("myWave1").Start tlStartImmediately

TheHdw.VHFACSource.Pins(SrcPin).Signals("myWave2").Start tlStartAtEnd
```

Comments

This statement:

.GainHardware.Mode = tlCSignalModeUseCalculatedValue tells the hardware to use the best setting for “GainHW” such that the programmed amplitude is realized with SMEM samples normalized.

This statement:

.GainHardware.Mode = tlCSignalModeUseLoadedValue

Tells the hardware to use the currently loaded setting for “GainHW” such that the programmed amplitude is realized by scaling SMEM.

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VhfacSource_theory.2.01

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VHFAC Source Theory of Operation

VHFAC Source Theory of Operation

The VHFAC Source instrument can generate high-accuracy, low-noise, and low-distortion signals with a bandwidth of 160MHz. It has a maximum user-programmable sampling rate of 400Msps and can drive a peak differential output of 4 V with DC coupling.

This section provides conceptual and reference information about the VHFAC Source hardware, features, and settings. The following topics are available:

- VHFAC Source Functional Description
- VHFAC Source Modes and Settings
- VHFAC Source Supported Modes Summary

For information on VHFAC Source instrument performance specifications and standards, see VHFAC Specifications.

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VhfacSource_theory.2.02

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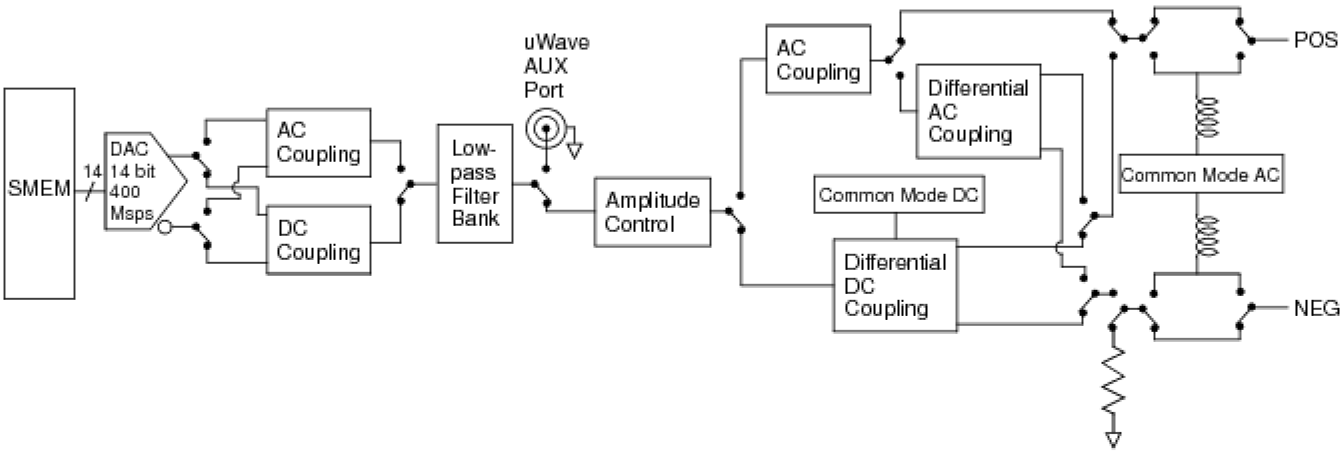
VHFAC Source Theory of Operation : VHFAC Source Functional Description

VHFAC Source Functional Description

The primary function of the VHFAC Source instrument is to provide analog signals of a specific shape, frequency, and amplitude to the device under test (DUT). This figure shows how the VHFAC Source instrument hardware generates and delivers an analog signal.



VHFAC Source Diagram

Details of each step in the process are described in the following topics:

- Source Memory
- Digital-to-Analog Conversion
- AC or DC Coupling
- Filtering

- AUX Port
- Amplitude Control
- Common Mode Offset
- Signal Delivery
- VHFAC Source Supported Modes Summary

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VHFAC Source Theory of Operation : VHFAC Source Functional Description : Source Memory

Source Memory

Normalized digital samples of the waveform are loaded in the source memory. The VHFAC Source memory is 8Msamples deep. However, the largest single segment length is 1Msample, allowing eight nesting levels. Nesting is used with compound segments. IG-XL automatically loads source segments as required by the start language and pattern starts.

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VHFAC Source Theory of Operation : VHFAC Source Functional Description : Digital-to-Analog Conversion

Digital-to-Analog Conversion

After leaving the source memory, the VHFAC Source instrument’s digital-to-analog converter (DAC) converts digital samples to an analog signal at a user-specified sample rate. The minimum user-supplied sample rate is 3.4ksps, and the maximum sample rate is 400Msps.

The Mixed Signal Workshop (MSW) tool provides timing solutions that meet the instrumentation sample rate requirements.

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VHFAC Source Theory of Operation : VHFAC Source Functional Description : AC or DC Coupling

AC or DC Coupling

The differential analog signal is then converted to single-ended, and AC or DC coupling is applied based on which one is selected by the test program. When AC coupled, the DC (low frequency) component is removed by circuitry in the analog front end. In AC coupled mode, the VHFAC Source can produce signals from 5 MHz to 160 MHz. In DC coupled mode, the VHFAC Source can produce signals from DC to 80 MHz.

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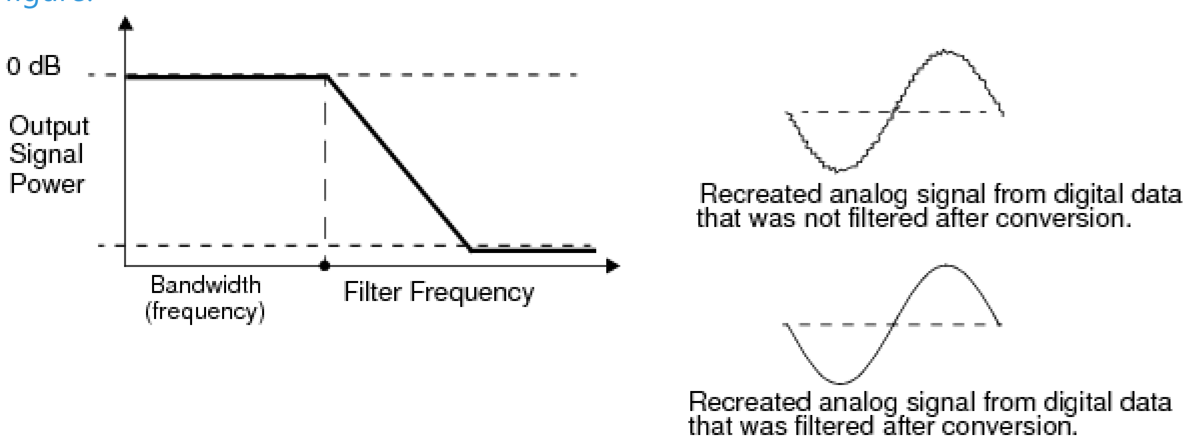
VHFAC Source Theory of Operation : VHFAC Source Functional Description : Filtering

Filtering

After passing through either AC or DC Coupling, the now single-ended analog signal passes through a user-selected filter or bypasses the filter altogether. This filter removes high frequency quantization noise and aliasing from the synthesized signal, and it smooths the sequence of discrete analog levels into a continuous analog signal. Multiple low-pass filters are available and when matched with the correct DAC sample rate produces a high-fidelity signal as shown in the figure.



Using a Filter to Reduce High Frequency Quantization Noise

See VHFAC Specifications for filter specifications.

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VHFAC Source Theory of Operation : VHFAC Source Functional Description : AUX Port

AUX Port

The single-ended analog signal can be optionally routed to the AUX port which is an auxiliary output path.

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VHFAC Source Theory of Operation : VHFAC Source Functional Description : Amplitude Control

Amplitude Control

After filtering, the analog signal passes through a voltage ranging stage, to attenuate the signal to the desired amplitude.

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VHFAC Source Theory of Operation : VHFAC Source Functional Description : Common Mode Offset

Common Mode Offset

After the amplitude is attenuated in the amplitude control stage, the analog signal is then AC or DC coupled and converted to single-ended or differential. DC coupled path common mode offset can be added to the source signal using the DC baseline circuitry in this stage (Common Mode DC). AC coupled path common mode offset can be added to the source signal using the DC baseline circuitry in this stage (Common Mode AC).

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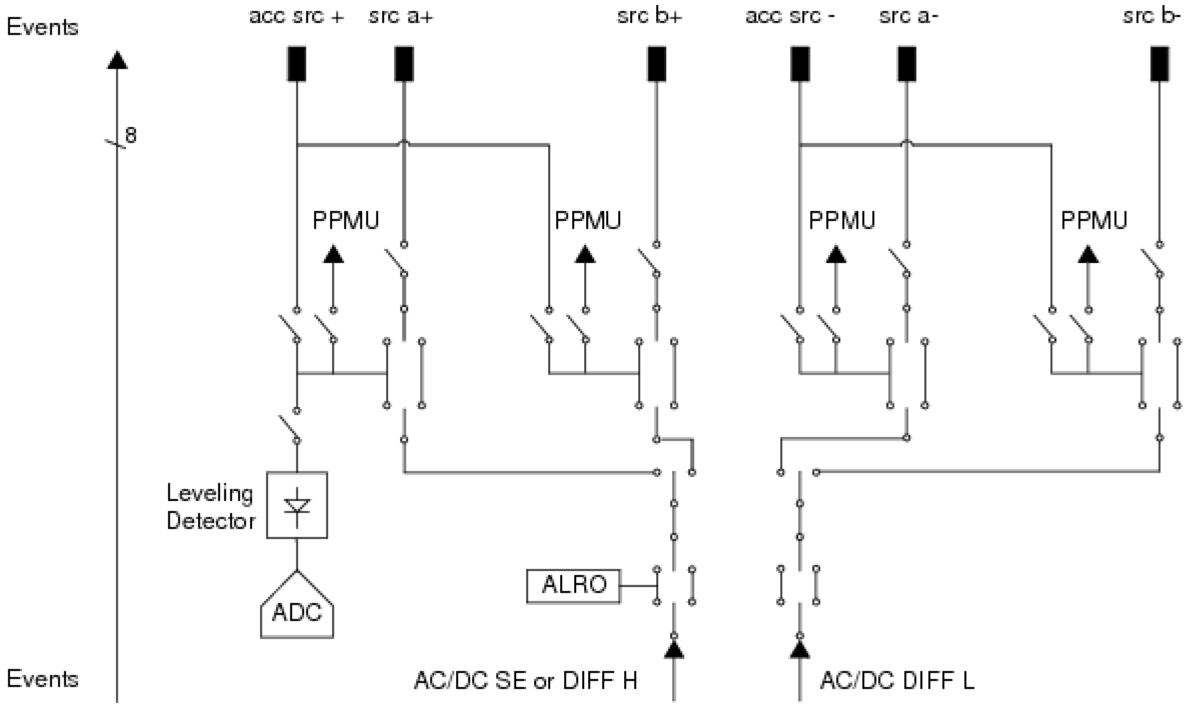
VHFAC Source Theory of Operation : VHFAC Source Functional Description : Signal Delivery

Signal Delivery

The VHFAC Source has a multiplexed output, allowing you to send signals to the DIB through either a “primary” or “secondary” differential signal path. The primary differential signals are denoted as “VHFACSourcePrimaryPos” (src a+) and “VHFACSourcePrimaryNeg” (src a-). The secondary differential signals are denoted as “VHFACSourceSecondaryPos” (src b+) and “VHFACSourceSecondaryNeg” (src b-).



Source Signal Delivery

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VHFAC Source Theory of Operation : VHFAC Source Modes and Settings

VHFAC Source Modes and Settings

The following topics introduce the concepts and features of some of the VHFAC Source instrument's operating modes and settings.

- Differential and Single-Ended Measurements
- VHFAC Source Reset Values

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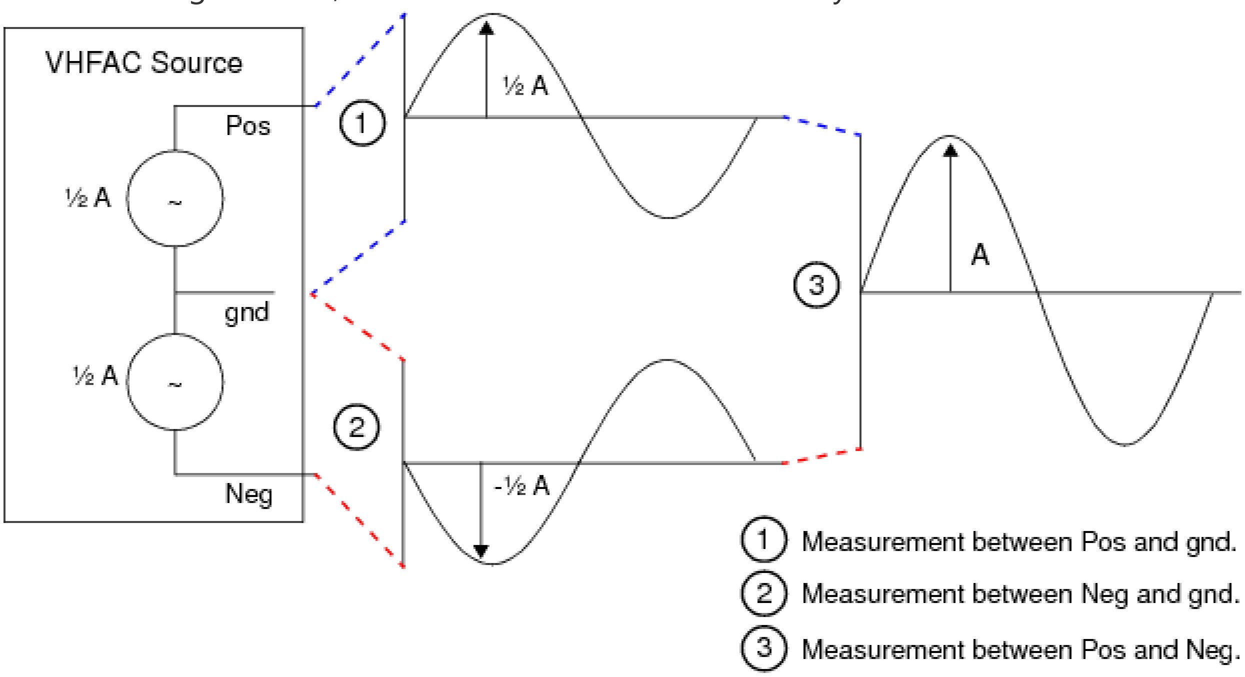
VHFAC Source Theory of Operation : VHFAC Source Modes and Settings : Differential and Single-Ended Measurements

Differential and Single-Ended Measurements

This figure shows a graphical representation of the VHFAC Source instrument where A is the programmed amplitude. The amplitude represents the differential voltage, measured between the positive (Pos) and negative (Neg) channels. Therefore, we say that the peak voltage (Vpk) is A when measured single-ended, and A when measured differentially.

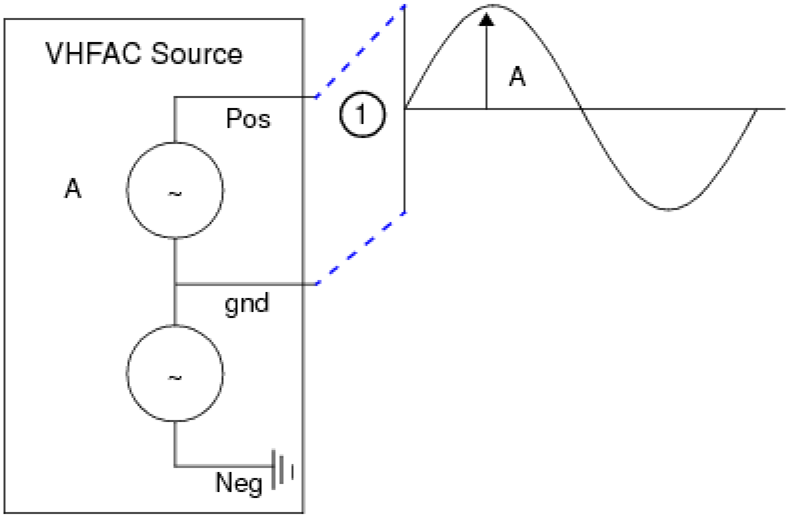


Differential Range Measurements

Likewise, the peak-to-peak voltage (Vpp) is defined as the maximum voltage minus the minimum voltage over time with respect to ground, which is A when measured single-ended and A when measured differentially (that is, each of the differential outputs (pos and neg) will be 1/2 the single-ended value).

VHFAC has two output modes which are single-ended and differential. In single-ended mode, the output signal is referenced from GND. In differential mode, the output signal is the voltage between Vpos and Vneg.

For single-ended mode, Vneg is terminated inside of the instrument and its output is disconnected.



Single-Ended Range Measurements

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VHFAC Source Theory of Operation : VHFAC Source Modes and Settings : VHFAC Source Reset Values

VHFAC Source Reset Values

This table lists the local reset values for the VHFAC Source. Pre-job reset, post-job reset, power-up reset, and user reset all result in the same state, unless otherwise noted.

VHFAC Reset Values

Settings	Default Value
Amplitude	1 Vpk
Low-Pass Filter	20 MHz
Low-Pass Filter Bypass	True
Output Type	SingleEnded
Coupling	DC
Load Impedance	Real = 50 ohm, Imaginary = 0
Enable Condition Bit	False
Connect Condition To	Both (board and PatGen)
AC Runtime Calibration Mode	Automatically
Calibration Frequency	5 MHz
CommonMode for DC	0V
Enable AC Path	False
CommonMode for AC	0V

Sample Rate	100 MHz
Sample Rate Enable	False

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VHFAC Source Theory of Operation : VHFAC Source Supported Modes Summary

VHFAC Source Supported Modes Summary

The following table summarizes the VHFAC Source modes.

VHFAC Source Supported Modes

			Segment Size Requirements[1]				
Output Type	Coupling	Signal Frequency Range (Fi)	1:1 Mux Mode 3.434kHz <= Fs <= 100MHz	2:1 Mux Mode 100MHz <= Fs <= 200MHz	4:1 Mux Mode 200MHz <= Fs <= 400MHz	Amplitude Vpk (50 ohm load)	Common Mode DC (50 ohm load)
Single-ended	DC	DC to 80MHz	35 - 1048576 samples Sample number must be modulo 1	70 - 1048576 samples Sample number must be modulo 2	140 - 1048576 samples Sample number must be modulo 4[2]	8.913mV - 2.0V	2.0 V
Differential						17.825mV - 4.0V	2.0 V
Single-ended	AC	5MHz to 160MHz				4.456mV - 1.0V	2.0 V
Differential						8.931mV - 2.0V	2.0 V

1 For compound segments, each “base segment” must follow these minimum sample size rules.

2 When using “UseLoadedValue” for the VHFAC Source sample rate, you must obey the >140 sample size rule.

If all segments that make up a compound segment comply with the modulo restrictions, then you can load more than the 1 Msample. However, you must follow certain rules.
For example, for sample rates between 100 MHz and 200 MHz, the modulo-2 restriction applies as follows.

- If the total sample size for a compound segment is <= 1Msample (1048576), the modulo-2 restriction applies to the compound segment.

- If the total sample size for a compound segment is greater than 1Msample (1048576), the modulo-2 restriction applies to each segment that makes up the compound segment.

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